

DiffGR: A Discrete Diffusion-Based Model for Personalised Recommendation by Reconstructing User-Item Bipartite Graphs

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Abstract. Generative models have recently been used to produce personalized recommendations by simulating user preferences in continuous space. However, existing work often overlooks crucial discrete structural relationships, such as graph sparsity and user-item topology, reducing recommendation accuracy. Current graph diffusion models that manipulate the full adjacency matrix in continuous space face scalability issues and poorly structured intermediates adjacency matrix. To address these problems, we develop DiffGR, a novel discrete diffusion model for personalised recommendation. DiffGR operates in discrete space, manipulating individual user-item bipartite sub-graphs by randomly removing edges in forward diffusion and restoring them based on a Bernoulli distribution in reverse, preserving graph structure and avoiding full adjacency matrix loading. Empirical studies show that DiffGR achieves competitive personalised recommendation results and generates synthetic graphs with high fidelity to the original dataset, suggesting a promising direction for discrete diffusion modelling in recommender systems.

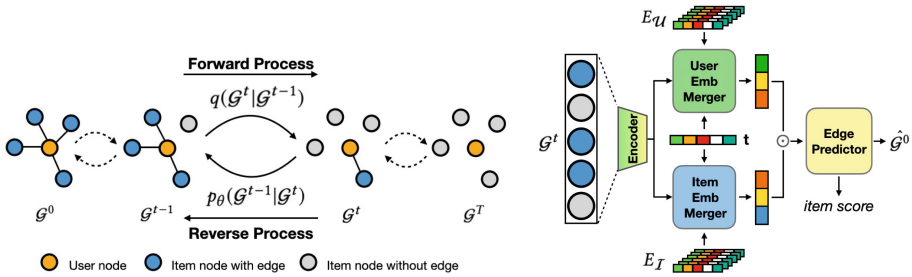
Keywords: Recommender Systems · Diffusion Models · Graph Generation

1 Introduction

Generative models including Generative Adversarial Networks (GANs) [20] and Variational Auto-Encoders (VAEs) [13] have demonstrated effectiveness in recommender systems (RSs) [21]. Generative recommenders are trained to predict the likelihood of user interactions with items they have not engaged with by learning the underlying generative process [20]. Recently, diffusion models (DMs) have gained more attention as state-of-the-art generative models in RS [21].

Limited research has explored the usage of DMs in RS. DMs [11,12,19,21] create personalised recommendations by gradually injecting noise into user profiles to simulate changes in interest and then eliminating the noise to align the new interests with the original preferences. However, current DMs in RS face

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(a) The diffusion process of DiffGR. (b) Denoising model structure.

Fig. 1. Overview of DiffGR: (a) edges in the user-item subgraphs are gradually removed and then restored; (b) the denoising model reconstructs the original graph through edge predictions.

significant limitations, often overlook the complex user-item relationships by operating solely in continuous space (e.g., latent representations) [23].

Most RS data possesses an inherently graphical structure [23,24]. Framing recommendation as a graph-based task effectively captures and uses the structural information that encompasses complex relationships between users and items within datasets [4]. Existing DMs [2,15,18] used for graph reconstruction often require the full adjacency matrix or extensive node and edge features, rendering them impractical for large-scale datasets commonly used in RSs.

To overcome these limitations, we propose DiffGR, the first diffusion-based recommender that operates the diffusion process directly on RS graphs in discrete space. DiffGR models the interactions of each user with items as a bipartite sub-graph, avoiding the computational need for the full adjacency matrix. During forward diffusion, edges are randomly removed in each sub-graph, and in the reverse process, they are restored following the Bernoulli distribution. By directly manipulating the graph in discrete space, DiffGR preserves the graph's structure, making it well-suited to large RS datasets. We conduct extensive experiments on three widely used datasets, comparing DiffGR with multiple competitive baselines. DiffGR achieves superior performance in personalized recommendation, demonstrating an effective use of discrete diffusion models in RSs. Its ability to generate high-fidelity synthetic graphs makes it a powerful tool for future research on generative RS models.

2 Discrete Diffusion

To preserve graph topology and capture user-item structural relationships, DiffGR is inspired by [2] and leverages the generation ability of diffusion models in discrete space for better personalised recommendations. Figure 1a illustrates the diffusion process of DiffGR. Each user's item interactions form a bipartite subgraph. The forward process randomly removes edges using Bernoulli noise until the graph is empty, while the reverse incrementally reconstructs it in discrete

space via a customized edge prediction model that follows a Bernoulli distribution. Output probabilities from this model are used to rank and recommend unseen items to users.

2.1 Discrete Forward and Reverse Process

Given a user-item bipartite sub-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i\}_{i=1}^{n}$ and $\mathcal{E} = \{e_i\}_{i=1}^{m}$ are the node set and edge set, respectively. $\mathcal{G}^0$ denotes the original sub-graph at time step $t = 0$. The diffusion process gradually corrupts $\mathcal{G}^0$ to an empty graph in T steps. The one-step diffusion transition probability $q(\mathcal{G}^t|\mathcal{G}^{t-1})$ from $\mathcal{G}^{t-1}$ to $\mathcal{G}^t$ can be formalized as:

$$q(\mathcal{G}^t|\mathcal{G}^{t-1}) = \prod_{i=1}^{m} \mathcal{B}(e_i^t; (1-\beta_t)e_i^{t-1}) \tag{1}$$

where $\mathcal{B}(e;\mu)$ represents the independent Bernoulli distribution where $e = 1$ with probability μ and $e = 0$ with probability $1-\mu$; β_t is the diffusion rate that determines the probability of e_i^{t-1} being removed from $\mathcal{G}^t$. Since these distributions form a Markov chain, the transition probability of any $\mathcal{G}^t$ given $\mathcal{G}^0$ can be represented as:

$$q(\mathcal{G}^t|\mathcal{G}^0) = \prod_{i=1}^{m} \mathcal{B}(e_i^t; \bar{\alpha}_t e_i^0). \tag{2}$$

where cumulative diffusion rates $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{t'=1}^{t} \alpha_{t'}$.

Due to the discrete nature of graphs, they are highly sensitive to added noise. A linear noise scheduler can quickly distort the graph in the early steps, making it difficult for the denoising model to learn the sampled distribution. To address this issue, a cosine noise scheduler is employed with a minor modification that what was previously $\sqrt{\bar{\alpha}_t}$ in continuous diffusion [14] becomes $\bar{\alpha}_t$ in discrete diffusion as suggested in [9]. This scheduler smoothly samples a small amount of noise in the early steps and follows a linear pattern in the middle stage. The cumulative diffusion rates $\bar{\alpha}_t$ are no longer obtained from the predefined noise rate β_t but directly defined using:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)} \quad f(t) = \cos\left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2}\right) \tag{3}$$

where s is a tunable hyper-parameter controlling the noise smoothness added to $\mathcal{G}^0$ in early steps. β_t is now noted as $\beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}$. Importance sampling is used to sample t for variance reduction. In the reverse process, a denoising model θ is trained to approximate $p_\theta(\mathcal{G}^{t-1}|\mathcal{G}^t)$ to the posterior of the forward diffusion process:

$$q(\mathcal{G}^{t-1}|\mathcal{G}^t, \mathcal{G}^0) = \prod_{i=1}^{m} \mathcal{B}\left(e_i^{t-1}; \frac{(1-\beta_t)e_i^{t-1} \cdot \bar{\alpha}_{t-1} e_i^0}{\bar{\alpha}_t e_i^0}\right) \tag{4}$$

Due to the large number of independent Bernoulli variables for potential edges in a user-item bipartite sub-graph, directly predicting noise in discrete

space is challenging. DiffGR predicts a probability vector for each edge $\hat{e_i^0}$ from the given e_i^t and then obtains $p_\theta(\mathcal{G}^{t-1}|\mathcal{G}^t)$ by calculating the posterior equation $\prod_{i=1}^{m} q(e_i^{t-1}|e_i^t, \hat{e_i^0})$. The clean graph $\mathcal{G}^0$ is approximated by the denoising network $\hat{\mathcal{G}^0} = \theta(\mathcal{G}^t, t)$. The denoising model θ is optimised by maximising the ELBO [8] of the clean user-item bipartite sub-graph $\mathcal{G}^0$:

$$\begin{aligned}\log p(\mathcal{G}^0) &= \mathbb{E}_{q(\mathcal{G}^{1:T}|\mathcal{G}^0)}\left[\log \frac{p_\theta(\mathcal{G}^{0:T})}{q(\mathcal{G}^{1:T}|\mathcal{G}^0)}\right] \\ &\geq \underbrace{\mathbb{E}_{q(\mathcal{G}^1|\mathcal{G}^0)}[\log p_\theta(\mathcal{G}^0|\mathcal{G}^1)]}_{\text{reconstruction term } \mathcal{L}_{rec}} - \underbrace{D_{KL}(q(\mathcal{G}^T|\mathcal{G}^0 \;||\; p(\mathcal{G}^T))}_{\text{prior matching term}} \\ &- \underbrace{\sum_{t=2}^{T} \mathbb{E}_{q(\mathcal{G}^t|\mathcal{G}^0)}[D_{KL}(q(\mathcal{G}^{t-1}|\mathcal{G}^t, \mathcal{G}^0) \;||\; p_\theta(\mathcal{G}^{t-1}|\mathcal{G}^t))]}_{\text{edge prediction term } \mathcal{L}_{edge}}\end{aligned} \tag{5}$$

where D_{KL} represents the KL-Divergence; D_{KL} in $\mathcal{L}_{edge}$ can be calculated as $D_{KL}(q(\mathcal{G}^{t-1}|\mathcal{G}^t, \mathcal{G}^0) \;||\; q(\mathcal{G}^{t-1}|\mathcal{G}^t, \hat{\mathcal{G}^0}))$, since DiffGR parametrizes $p_\theta(\mathcal{G}^{t-1}|\mathcal{G}^t)$ using the approximation $\hat{\mathcal{G}^0}$ and the posterior equation of q [9]. The prior matching term is eliminated during the training process as it is a constant. The reconstruction term $\mathcal{L}_{rec}$ can be computed as:

$$\mathcal{L}_{rec} = \sum_{i=1}^{m} e_{i,b}^0 \log \hat{e}_{i,b}^0, \quad b \in \{0, 1\} \tag{6}$$

2.2 Personalised Recommendation and Graph Generation

Our denoising model has four modules (Fig. 1b), reconstructing the edges through its approximation $\hat{\mathcal{G}^0}$ given the noise input $\mathcal{G}^t$. An encoder first learns the latent representation $emb_{\mathcal{G}^t}$ of $\mathcal{G}^t$. As the diffusion works on individual user-item sub-graphs, mutual information across sub-graphs is hard to capture. To counter this, we use an ID embedding module projecting node IDs into embeddings (e.g., $Emb_{\mathcal{U}}$ and $Emb_{\mathcal{I}}$ for user and item nodes respectively). For a sub-graph $\mathcal{G}$, the user embedding merger concatenates emb_u, $emb_{\mathcal{G}^t}$ and time step embedding emb_t, producing emb_{final_u}. We encode emb_t using Sinusoidal positional embedding [3]. Similarly, the entire item node embeddings $Emb_{final_{\mathcal{I}}}$ are produced. Edge embeddings Emb_e are formed via the Hadamard product between emb_{final_u} and $Emb_{final_{\mathcal{I}}}$ and sent to the edge prediction module to predict each edge's Bernoulli probability $\hat{e}_i^0$ of $\hat{\mathcal{G}^0}$. The ID embedding module injects mutual information across different bipartite graphs, enriching edge embeddings.

In inference, we rebuild $\mathcal{G}^0$ from a corrupted graph $\mathcal{G}^T$, as the denoising model has already learnt the personalised information. At time step $t = 0$, each edge's Bernoulli probability $\hat{e}_i^0$ is processed by a $softmax$ function so that the probability of being 0 (no edge) and 1 (exists an edge) sums to 1, and each user-item pair's score is calculated as follows:

$$score_i = \hat{e}_{i_1}^0 - \hat{e}_{i_0}^0. \tag{7}$$

The scores are ranked and used to produce personalised recommendations. The user-item bipartite sub-graph $\mathcal{G}^0$ is reconstructed by eliminating the edges with a high probability of being 0 from $\hat{\mathcal{G}}^0$.

3 Experiments Setup

Table 1. Overall performance comparison. Best results in bold, second-best underlined. %Improve. indicates DiffGR's improvement over the top baseline.

Method	ML-1M		Pinterest		Yelp2018	
	R@20	N@20	R@20	N@20	R@20	N@20
BPR	0.1793	0.2624	0.1053	0.0710	0.0585	0.0485
LightGCN	0.1731	0.3030	0.1301	0.0892	0.0751	0.0645
MultiVAE	<u>0.2312</u>	0.3177	0.1261	0.0850	0.0789	0.0664
CODIGEM	0.2221	0.3211	0.1291	0.0898	0.0552	0.0476
DiffRec	0.2280	<u>0.3287</u>	0.1318	0.0915	0.0787	<u>0.0697</u>
L-DiffRec	0.2247	0.3111	<u>0.1384</u>	<u>0.0930</u>	<u>0.0802</u>	0.0667
DiffGR	**0.2397**	**0.3413**	**0.1445**	**0.0989**	**0.0844**	**0.0730**
%Improve.	3.68%	3.83%	4.41%	6.34%	5.24%	4.73%

We conduct experiments on three widely used datasets: 1) **ML-1M**[1] (movie ratings with 22.4M edges), 2) **Pinterest** [6] (user-pinned images with 0.53B edges), and 3) **Yelp2018** [22] (restaurant reviews with 1.2B edges). User interactions are randomly split into 70% train, 10% validation, and 20% test sets. To preserve real user behavior, we use minimal pre-processing [5,17], excluding only users with fewer than five interactions for sufficient training data and accurate graph reconstruction in reverse diffusion.

We compare DiffGR with six competitive and well-known baselines, covering: pairwise learning to rank **BPR** [16], light graph convolution **LightGCN** [7], generative VAE **Mult-VAE** [13] and continuous denoising diffusion-based recommenders **CODIGEM** [19], **DiffRec** [21], and **L-DiffRec** [21].

Parameters for all approaches are tuned following [21], except for L-DiffRec. For L-DiffRec, we set item clusters to 1, use default VAE settings, and tune other parameters the same as DiffRec. For DiffGR, we tune the cosine noise scheduler offset s from $\{0.8, 0.08, 0.008\}$ and select dropout ratio in $\{0.1, 0.3, 0.5\}$. Diffusion steps are set to $T = 5$, and embedding mergers and encoder structures are fixed for efficiency. All models are trained with learning rates from $\{1e^{-2}, 1e^{-3}, 1e^{-4}\}$.

[1] https://grouplens.org/datasets/movielens/1m/.

Table 2. Statistics of generated graphs.

Dataset	User		Item	
	$\rho \uparrow$	PLE $\downarrow$	$\rho \uparrow$	PLE $\downarrow$
Ml-1M	0.9868	0.0120	0.9949	0.0429
Pinterest	0.6711	0.2350	0.9788	0.1910
Yelp2018	0.8569	0.0161	0.8884	0.0504

The recommendation performance is evaluated using *Recall@K* ($R@K$) and $NDCG@K$ ($N@K$) [10], where K is set to 20. The Spearman Rank-order Correlation (ρ) [26] assesses the node positions between synthesized and original graphs. The Power-Law Exponent (PLE) [1] evaluates the distribution distance between them.

4 Experiment Results

Table 1 reports each algorithm's recommendation performance by dataset. DiffGR outperforms both generative and non-generative models across all datasets, demonstrating that it benefits from modelling the complex structural of user-item graphs through the diffusion process in discrete space. It shows stronger gains on sparser datasets (e.g., **Pinterest** (sparsity: 99.73%) and **Yelp2018** (99.98%) compared to ML-1M (95.53%)), where traditional and continuous diffusion models may struggle with complex, implicit user preference. DiffGR handles sparse graphs at each time step and is naturally more capable of learning from extremely sparse datasets, demonstrating the advantages of discrete diffusion models in RSs over other models.

To further analyse the synthetic graph generated by DiffGR, we compare its hub-like structure and power-law distribution to the original graph using (ρ) [26] and (PLE) [1], respectively. The original and generated user and item nodes are ranked by edge count in both graphs, with ρ measuring the correlation between these rankings. As shown in Table 2, ρ values for users and items are close to 1 (mostly > 0.9), indicating that nodes with higher edge counts in the original graph retain them in the generated graph. This preserves node positioning similarity, with hub-center nodes remaining central. The PLE indicates how node degree frequencies in the synthetic graph follow the original graph's power-law distribution. Table 2 shows low PLE values (around 0.05, except for **Pinterest**), suggesting that the generated graph closely preserves this distribution. Figure 2 shows the distribution of embeddings learned by LightGCN of original and generated users. Both distributions are similar, with only minor angular shifts due to diffusion process uncertainty.

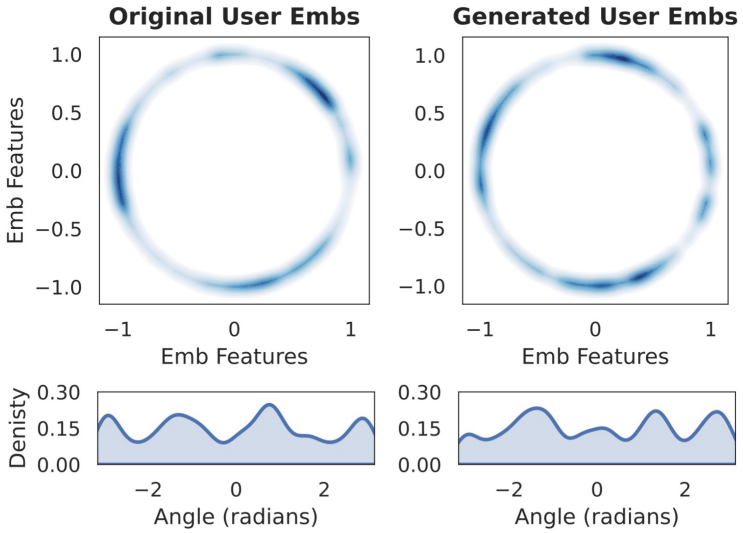


Fig. 2. The visualization of user embedding distributions on Yelp2018, following the approach of [25]. Clusters are represented by areas in darker colours. Distribution shifts are shown by the density of angles.

5 Conclusion and Future Work

In this paper, we present DiffGR, the first discrete diffusion-based generative RS. DiffGR focuses on modelling the crucial structural relationships between users and items in a discrete space by diffusing user-item graphs. Moreover, it avoids the need for loading entire adjacency matrices by directly manipulating each user's interaction graphs and we show the generated graphs match the properties of the original. The experimental results show that DiffGR achieves state of the art performance. We expect DiffGR's ability to generate synthetic graphs will stimulate interesting new directions in generative modelling for RSs.

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References

1. Alstott, J., Bullmore, E., Plenz, D.: powerlaw: A python package for analysis of heavy-tailed distributions. PLoS ONE 9(1), e85777 (2014). https://doi.org/10.1371/journal.pone.0085777, http://dx.doi.org/10.1371/journal.pone.0085777
2. Chen, X., He, J., Han, X., Liu, L.: Efficient and degree-guided graph generation via discrete diffusion modeling. In: International Conference on Machine Learning, pp. 4585–4610. PMLR (2023)

3. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding (2019). https://arxiv.org/abs/1810.04805
4. Fan, W., Ma, Y., Li, Q., He, Y., Zhao, E., Tang, J., Yin, D.: Graph neural networks for social recommendation. In: The World Wide Web Conference, pp. 417–426 (2019)
5. Ferrari Dacrema, M., Cremonesi, P., Jannach, D.: Are we really making much progress? a worrying analysis of recent neural recommendation approaches. In: Proceedings of the 13th ACM Conference on Recommender Systems, pp. 101–109 (2019)
6. Geng, X., Zhang, H., Bian, J., Chua, T.S.: Learning image and user features for recommendation in social networks. In: Proceedings of the IEEE international conference on computer vision. pp. 4274–4282 (2015)
7. He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M.: Lightgcn: Simplifying and powering graph convolution network for recommendation. In: Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, pp. 639–648 (2020)
8. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models (2020)
9. Hoogeboom, E., Nielsen, D., Jaini, P., Forré, P., Welling, M.: Argmax flows and multinomial diffusion: Learning categorical distributions. Adv. Neural. Inf. Process. Syst. **34**, 12454–12465 (2021)
10. Järvelin, K., Kekäläinen, J.: Ir evaluation methods for retrieving highly relevant documents. In: ACM SIGIR Forum, vol. 51, pp. 243–250. ACM, New York (2017)
11. Jiang, Y., Yang, Y., Xia, L., Huang, C.: Diffkg: knowledge graph diffusion model for recommendation. In: Proceedings of the 17th ACM International Conference on Web Search and Data Mining, pp. 313–321 (2024)
12. Li, Z., Sun, A., Li, C.: Diffurec: a diffusion model for sequential recommendation. ACM Trans. Inf. Syst. **42**(3), 1–28 (2023)
13. Liang, D., Krishnan, R.G., Hoffman, M.D., Jebara, T.: Variational autoencoders for collaborative filtering. In: Proceedings of the 2018 World Wide Web Conference, pp. 689–698 (2018)
14. Nichol, A.Q., Dhariwal, P.: Improved denoising diffusion probabilistic models. In: International Conference on Machine Learning, pp. 8162–8171. PMLR (2021)
15. Niu, C., Song, Y., Song, J., Zhao, S., Grover, A., Ermon, S.: Permutation invariant graph generation via score-based generative modeling. In: International Conference on Artificial Intelligence and Statistics, pp. 4474–4484. PMLR (2020)
16. Rendle, S., Freudenthaler, C., Gantner, Z., Schmidt-Thieme, L.: Bpr: bayesian personalized ranking from implicit feedback. In: Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence, pp. 452–461. UAI '09. AUAI Press, Arlington (2009)
17. Sachdeva, N., McAuley, J.: How useful are reviews for recommendation? a critical review and potential improvements. In: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 1845–1848 (2020)
18. Vignac, C., Krawczuk, I., Siraudin, A., Wang, B., Cevher, V., Frossard, P.: Digress: discrete denoising diffusion for graph generation. arXiv preprint arXiv:2209.14734 (2022)
19. Walker, J., Zhong, T., Zhang, F., Gao, Q., Zhou, F.: Recommendation via collaborative diffusion generative model. In: International Conference on Knowledge Science, Engineering and Management, pp. 593–605. Springer (2022)

20. Wang, J., Yu, L., Zhang, W., Gong, Y., Xu, Y., Wang, B., Zhang, P., Zhang, D.: Irgan: a minimax game for unifying generative and discriminative information retrieval models. In: Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 515–524 (2017)
21. Wang, W., Xu, Y., Feng, F., Lin, X., He, X., Chua, T.S.: Diffusion recommender model. In: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 832–841 (2023)
22. Wang, X., He, X., Wang, M., Feng, F., Chua, T.S.: Neural graph collaborative filtering. In: Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval. SIGIR '19. ACM, July 2019. https://doi.org/10.1145/3331184.3331267
23. Wu, S., Sun, F., Zhang, W., Xie, X., Cui, B.: Graph neural networks in recommender systems: a survey. ACM Comput. Surv. **55**(5), 1–37 (2022)
24. Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W.L., Leskovec, J.: Graph convolutional neural networks for web-scale recommender systems. In: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 974–983 (2018)
25. Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., Nguyen, Q.V.H.: Are graph augmentations necessary? simple graph contrastive learning for recommendation (2022)
26. Zar, J.H.: Spearman rank correlation. Encyclopedia of biostatistics **7** (2005)